



Statistics and Data Science

StataNow 19.5
BE-*Basic Edition*

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Notes:

1. Unicode is supported; see [help unicode advice](#).
2. New update available; type `-update all-`

```
1 . doedit "C:\Users\yoaga\OneDrive\Escritorio\Survival\PruebaFinal_2\CodigoFinal_Rev.do"
2 . clear
3 .
4 . use amdata.dta
5 .
6 . count
   85
7 .
8 .
9 .
10 . generate t = recdate-dxdate
    (41 missing values generated)
11 .
12 . replace t = lndate-dxdate if recdate == .
    (41 real changes made)
13 .
14 .
15 .
16 . generate cstage = stage
17 .
18 . destring cstage, replace ignore("NA")
    cstage already numeric; no replace
19 .
20 . replace cstage = . if cstage == 0
    (1 real change made, 1 to missing)
```

```

21 .
22 . replace cstage = min(cstage,3)
    (6 real changes made)

23 .
24 .
25 .
26 . replace cstage = max(cstage,2)
    (10 real changes made)

27 .
28 . destring cluster, replace ignore("NA")
    cluster: characters N A removed; replaced as byte
    (41 missing values generated)

29 .
30 . tab imut recurrence, all row

```

Key
<i>frequency</i>
<i>row percentage</i>

imut	recurrence		Total
	0	1	
0	27	16	43
	62.79	37.21	100.00
1	14	28	42
	33.33	66.67	100.00
Total	41	44	85
	48.24	51.76	100.00

```

      Pearson chi2(1) = 7.3839 Pr = 0.007
Likelihood-ratio chi2(1) = 7.4967 Pr = 0.006
      Cramér's V = 0.2947
      gamma = 0.5429 ASE = 0.160
Kendall's tau-b = 0.2947 ASE = 0.104

```

```

31 .
32 .
33 .
34 . generate dxcat = .
    (85 missing values generated)

35 .
36 . replace dxcat = 0 if dxyear < 2014.5
    (19 real changes made)

37 .
38 . replace dxcat = 1 if dxyear > 2016.5
    (40 real changes made)

```

```
39 .
40 .
41 .
42 . xi: logistic imut dxdate sex age Q2AMR cstage
```

Logistic regression

Number of obs = 73
 LR chi2(5) = 13.16
 Prob > chi2 = 0.0219
 Pseudo R2 = 0.1305

Log likelihood = -43.849075

imut	Odds ratio	Std. err.	z	P> z	[95% conf. interval]	
dxdate	1.000281	.0003449	0.82	0.415	.9996054	1.000957
sex	.2619569	.1417737	-2.48	0.013	.0906883	.7566736
age	.9646099	.0228448	-1.52	0.128	.9208582	1.01044
Q2AMR	.0625818	.0849547	-2.04	0.041	.0043746	.8952685
cstage	1.597506	.8650562	0.87	0.387	.5527302	4.617126
_cons	.1060477	.7468898	-0.32	0.750	1.07e-07	104828.3

Note: _cons estimates baseline odds.

```
43 .
44 . tab imut dxcat, all row
```

Key
<i>frequency</i>
<i>row percentage</i>

imut	dxcat		Total
	0	1	
0	12 41.38	17 58.62	29 100.00
1	7 23.33	23 76.67	30 100.00
Total	19 32.20	40 67.80	59 100.00

Pearson chi2(1) = 2.1995 Pr = 0.138
 Likelihood-ratio chi2(1) = 2.2179 Pr = 0.136
 Cramér's V = 0.1931
 gamma = 0.3975 ASE = 0.241
 Kendall's tau-b = 0.1931 ASE = 0.127

```
45 .
46 .
47 .
48 . spearman dxdate imut
```

Number of observations = 85
 Spearman's rho = 0.1630

Test of H0: dxdate and imut are independent
 Prob = 0.1356

49 .
 50 . tab imut cstage, all row

Key
<i>frequency</i> <i>row percentage</i>

imut	cstage		Total
	2	3	
0	19	24	43
	44.19	55.81	100.00
1	15	27	42
	35.71	64.29	100.00
Total	34	51	85
	40.00	60.00	100.00

Pearson chi2(1) = 0.6354 Pr = 0.425
 Likelihood-ratio chi2(1) = 0.6365 Pr = 0.425
 Cramér's V = 0.0865
 gamma = 0.1753 ASE = 0.216
 Kendall's tau-b = 0.0865 ASE = 0.108

51 .
 52 .
 53 .
 54 . tab imut sex, all row col

Key
<i>frequency</i> <i>row percentage</i> <i>column percentage</i>

imut	sex		Total
	0	1	
0	18	25	43
	41.86	58.14	100.00
	36.73	69.44	50.59
1	31	11	42
	73.81	26.19	100.00
	63.27	30.56	49.41
Total	49	36	85
	57.65	42.35	100.00
	100.00	100.00	100.00

Pearson chi2(1) = 8.8829 Pr = 0.003
 Likelihood-ratio chi2(1) = 9.0696 Pr = 0.003
 Cramér's V = -0.3233
 gamma = -0.5930 ASE = 0.152
 Kendall's tau-b = -0.3233 ASE = 0.102

```
55 .
56 . tab imut recurrence, row all
```

Key
<i>frequency</i> <i>row percentage</i>

imut	recurrence		Total
	0	1	
0	27 62.79	16 37.21	43 100.00
1	14 33.33	28 66.67	42 100.00
Total	41 48.24	44 51.76	85 100.00

```

Pearson chi2(1) = 7.3839 Pr = 0.007
Likelihood-ratio chi2(1) = 7.4967 Pr = 0.006
Cramér's V = 0.2947
gamma = 0.5429 ASE = 0.160
Kendall's tau-b = 0.2947 ASE = 0.104
```

```
57 .
58 .
59 .
60 . tab mutation recurrence, row all exact
```

Key
<i>frequency</i> <i>row percentage</i>

Enumerating sample-space combinations:

```
stage 6: enumerations = 1
stage 5: enumerations = 2
stage 4: enumerations = 14
stage 3: enumerations = 88
stage 2: enumerations = 203
stage 1: enumerations = 0
```

Mutation_s tatus	recurrence		Total
	0	1	
braf	4 36.36	7 63.64	11 100.00
kit	4 36.36	7 63.64	11 100.00
multihit	0 0.00	1 100.00	1 100.00
nf1	0 0.00	7 100.00	7 100.00
nras	6 50.00	6 50.00	12 100.00
qwt	27	16	43

	62.79	37.21	100.00
Total	41	44	85
	48.24	51.76	100.00

Pearson chi2(5) = **12.3598** Pr = **0.030**
 Likelihood-ratio chi2(5) = **15.4872** Pr = **0.008**
 Cramér's V = **0.3813**
 gamma = **-0.4313** ASE = **0.148**
 Kendall's tau-b = **-0.2615** ASE = **0.095**
 Fisher's exact = **0.018**

61 .

62 . ltable t recurrence, test notable graph overlay xlab(0(400)4000) ylab(0(0.2)1) by(mutation) xtitle("Since recruitment
> roportion Relapse-free") ci

Likelihood-ratio test statistic of homogeneity (group=mutation_status):
 chi2(5) = **14.307129**, P = **.01377181**

Logrank test of homogeneity (group=mutation_status):

Equality of survivor functions
 Log-rank test

mutation_s~s	Observed events	Expected events
brf	7	6.29
kit	7	3.25
multihit	1	0.30
nf1	7	2.84
nrns	6	6.82
qwt	16	24.50
Total	44	44.00

chi2(5) = **15.37**
 Pr>chi2 = **0.0089**

63 .

64 . graph export ltable_recurrence_Mutation.pdf, as(pdf) replace
file ltable_recurrence_Mutation.pdf saved as PDF format

65 .

66 .

67 .

68 .

69 .

70 . stset t, failure(recurrence) origin(time dxdate) enter(time conddate)

Survival-time data settings

Failure event: **recurrence!=0 & recurrence<.**
 Observed time interval: **(origin, t]**
 Enter on or after: **time conddate**
 Exit on or before: **failure**
 Time for analysis: **(time-origin)**
 Origin: **time dxdate**

```

85 total observations
85 observations end on or before enter()

```

```

0 observations remaining, representing
0 failures in single-record/single-failure data
0 total analysis time at risk and under observation
      At risk from t =      0
      Earliest observed entry t =      .
      Last observed exit t =      .

```

```

71 .
72 .
73 .
74 .
75 .
76 . xi: logistic recurrence imut sex age cstage dxdate Q2AMR

```

```

Logistic regression                                Number of obs =    73
                                                    LR chi2(6)      =   18.11
                                                    Prob > chi2     =  0.0060
Log likelihood = -41.484649                        Pseudo R2      =  0.1791

```

recurrence	Odds ratio	Std. err.	z	P> z	[95% conf. interval]	
imut	5.310556	3.325634	2.67	0.008	1.556293	18.12127
sex	.9844121	.5578017	-0.03	0.978	.3242328	2.988801
age	1.016565	.0244662	0.68	0.495	.9697256	1.065667
cstage	3.545433	1.973259	2.27	0.023	1.191027	10.554
dxdate	1.000076	.000339	0.23	0.822	.9994121	1.000741
Q2AMR	21.06773	33.44283	1.92	0.055	.9384777	472.9461
_cons	.0001449	.0010369	-1.24	0.217	1.18e-10	178.0292

Note: **_cons** estimates baseline odds.

```

77 .
78 .
79 .
80 .
81 .
82 . xi: logistic recurrence i.cluster sex age i.cstage dxdate Q2AMR
i.cluster      _Icluster_1-3      (naturally coded; _Icluster_1 omitted)
i.cstage       _Icstage_2-3       (naturally coded; _Icstage_2 omitted)

```

```

Logistic regression                                Number of obs =    37
                                                    LR chi2(7)      =   10.93
                                                    Prob > chi2     =  0.1419
Log likelihood = -19.844933                        Pseudo R2      =  0.2158

```

recurrence	Odds ratio	Std. err.	z	P> z	[95% conf. interval]	
_Icluster_2	9.164577	12.29136	1.65	0.099	.6614502	126.9778
_Icluster_3	3.599498	3.73278	1.24	0.217	.4715438	27.47652
sex	1.139034	.9488292	0.16	0.876	.2225746	5.829051
age	.9679004	.0353427	-0.89	0.372	.9010507	1.03971
_Icstage_3	2.638177	2.134682	1.20	0.231	.5401993	12.88409
dxdate	1.000365	.0007233	0.50	0.614	.998948	1.001783
Q2AMR	133.2494	344.2276	1.89	0.058	.842808	21066.97
_cons	.0000293	.0004368	-0.70	0.484	5.93e-18	1.45e+08

Note: **_cons** estimates baseline odds.

```

83 .
84 .
85 .
86 . ltable t recurrence, test notable graph overlay xlab(0(400)4000) ylab(0(0.2)1) by(cluster) xtitle("Since recruitment (
> oportion Relapse-free") ci

```

Likelihood-ratio test statistic of homogeneity (group=cluster):
 $\chi^2(2) = 14.223938, \quad P = .00081529$

Logrank test of homogeneity (group=cluster):

Equality of survivor functions
 Log-rank test

cluster	Observed events	Expected events
1	5	11.68
2	13	6.28
3	8	8.04
Total	26	26.00

$\chi^2(2) = 11.64$
 $Pr>\chi^2 = 0.0030$

```

87 .
88 . graph export ltable_recurrence_Cluster.pdf, as(pdf) replace
    file ltable_recurrence_Cluster.pdf saved as PDF format
89 .
90 .
91 .
92 . tab cluster recurrence , exact row

```

Key
<i>frequency</i>
<i>row percentage</i>

Enumerating sample-space combinations:
 stage 3: enumerations = 1
 stage 2: enumerations = 8
 stage 1: enumerations = 0

RNA_cluste r	recurrence		Total
	0	1	
1	9 64.29	5 35.71	14 100.00
2	3 18.75	13 81.25	16 100.00
3	6 42.86	8 57.14	14 100.00
Total	18 40.91	26 59.09	44 100.00

Fisher's exact = 0.039

93 .
94 .
95 .
96 .
97 .
98 .
99 .
100 .
101 .
102 .
103 .
104 .
105 .